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COURSE CODE: 24SDCS02 / 24SDCS02E / 24SDCS02P / 24SDCS02L

FULL STACK APPLICATION DEVELOPMENT

Date of the Session: _/_/_____ Time of The Session: __to_____

#SKILL - 1 Git Version Control

Prerequisites:

- Basic knowledge of Git commands
- Ability to create and edit project files
- Understanding of version control basics

The development team wants to manage their project using Git to maintain a structured and collaborative workflow. In this lab, you will simulate a real-world version control process: creating project files, tracking and committing changes, using GitHub as remote storage, working with multiple branches, and handling merge conflicts. This helps you understand how developers work together efficiently in software projects.

Tasks:

1. Create a new project folder, initialize Git using git init, and configure Git with your username and email.
2. Create two files (example: main.java, notes.txt) with sample content, check the repository status, and add them to staging.
3. Commit the staged files with a meaningful commit message.
4. Create a new repository on GitHub, connect your local repo using the remote URL and push the committed changes.
5. Create the first branch (example: feature-update), make modifications, and add + commit the changes.
6. Create a second branch (example: bug-fix), apply changes, and add + commit them.
7. Switch back to the main branch and merge both branches one after the other.
8. If any merge conflict occurs, resolve it manually, commit the fix, and complete the merge.

Git commands:

```
vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ cd Desktop
bash: cd: Desktop: No such file or directory

vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ mkdir git-akill1-project

vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ cd git-skill1-project
bash: cd: git-skill1-project: No such file or directory

vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ cd ~

vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ pwd
/c/Users/vasav

vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ ls
 AppData/
'Application Data'@
'Cisco Packet Tracer 9.0.0'/
 Contacts/
 Cookies@
 Documents/
 Downloads/
 Favorites/
 Links/
'Local Settings'@
 MO1.1.ipynb
 MO1.ipynb
 Music/
'My Documents'@
 NTUSER.DAT
 NTUSER.DAT{6d38259f-776b-11f0-af47-f29f25244859}.TM.blf
 NTUSER.DAT{6d38259f-776b-11f0-af47-f29f25244859}.TMContainer00000000000000000001.regtrans-ms
 NTUSER.DAT{6d38259f-776b-11f0-af47-f29f25244859}.TMContainer00000000000000000002.regtrans-ms
 NetHood@
 OneDrive/
 PDFEditor/
 Postman/
 PrintHood@
 PyCharmMiscProject/
 PycharmProjects/
 Recent@
'Saved Games'/
 Searches/
 SendTo@
'Start Menu'@
 Templates@
 Videos/
 eclipse/
 eclipse-workspace/
 git-akill1-project/
 mentalmentor/
 ntuser.dat.LOG1
```

MINGW64:/c/Users/vasav/OneDrive/Desktop/git-skill1-project

```
vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ ^C

vasav@LAPTOP-08PIEVOB MINGW64 ~ (main)
$ cd OneDrive

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive (main)
$ ls
 AppData/   Insemexam/            Pictures/   desktop.ini   'oops sec212'/   文档/
 Desktop/  'Personal Vault.lnk'*  'SEC 207'/  'full stack'/   react/

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive (main)
$ cd Desktop

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop (main)
$ pwd
/c/Users/vasav/OneDrive/Desktop

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop (main)
$ mkdir git-skill1-project

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop (main)
$ cd git-skill1-project

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ pwd
/c/Users/vasav/OneDrive/Desktop/git-skill1-project

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ git init
Initialized empty Git repository in C:/Users/vasav/OneDrive/Desktop/git-skill1-project/.git/

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ touch main.java

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ touch notes.txt

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ ls
main.java  notes.txt

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ notepad main.java

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ notepad notes.txt

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        main.java
```

```
nothing added to commit but untracked files present (use "git add" to track)

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
        add

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        main.java
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ ^C

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
        add

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git add .

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   main.java
        new file:   notes.txt


vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git commit -m "Initial commit"
[master (root-commit) 81d0128] Initial commit
 2 files changed, 7 insertions(+)
 create mode 100644 main.java
 create mode 100644 notes.txt

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ ^C
```

```
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   main.java
        new file:   notes.txt


vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git commit -m "Initial commit"
[master (root-commit) 81d0128] Initial commit
 2 files changed, 7 insertions(+)
 create mode 100644 main.java
 create mode 100644 notes.txt

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ ^C

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git remote add origin https://github.com/Vasavo7575/FSAD-WORKBOOK.git

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (master)
$ git branch -M main

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ git push -u origin main
remote: Repository not found.
fatal: repository 'https://github.com/Vasavo7575/FSAD-WORKBOOK.git/' not found

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ ^C

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ git remote remove origin

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ git remote add origin https://github.com/Vasavi7575/FSAD-WORKBOOK.git

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 410 bytes | 410.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Vasavi7575/FSAD-WORKBOOK.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ ^C

vasav@LAPTOP-08PIEVOB MINGW64 ~/OneDrive/Desktop/git-skill1-project (main)
$ |
```

Main.java

```
public class Main {
  public static void main(String[] args) {
    System.out.println("Hello Git");
  }
}
```

Notes.txt

Git Skill 1 Lab

Basic version control practice

VIVA QUESTIONS:

1. Why is Git used in development?
2. What information is stored in Git configuration?
3. What is Maven and why is it used?
4. What is the role of the POM.xml file?
5. Explain Maven Build Life Cycle.

1Ans: Git is a **version control system** that helps developers manage changes to their code.
Reasons for using Git:

- Tracks the history of code changes.
- Allows multiple developers to **collaborate** on the same project.

2Ans: Git configuration stores settings that control Git behavior. There are three levels: **system**, **global**, and **local**.
Information stored includes:

- User identity: user.name and user.email
- Default text editor for commits

  You can view it with:

Command: git config –list

3Ans: Maven is a **build automation and dependency management tool** for Java projects.
Why it's used:

- Manages project **build process** (compilation, packaging, testing, deployment).
- Handles **dependencies** automatically (libraries required by the project).
- Supports **plugins** to extend functionality (e.g., generating reports, running tests).

4Ans: The **POM (Project Object Model) file** is the **core configuration file of a Maven project**.
It defines:

- Project information: name, version, description
- Dependencies needed by the project
- Source directories and packaging type (jar, war, etc.)

5Ans: Maven has **three built-in build life cycles**: **default**, **clean**, and **site**. The most important is **default**, which handles project build.

Phases of Default Build Life Cycle:

1. **validate** – Check if project is correct and necessary info is available.
2. **compile** – Compile the source code.
3. **test** – Run tests using a testing framework (e.g., JUnit).
4. **package** – Package compiled code into a JAR/WAR.
5. **verify** – Run checks to ensure the package is valid.
6. **install** – Install the package into the local Maven repository.
7. **deploy** – Deploy the package to a remote repository for sharing.